Computing Black-Hole Quasi-Normal Modes From a Physics-Informed Network to a Hybrid Neural-Operator Surrogate

Forward PINN and Hybrid FNO on Schwarzschild $\ell = 2$

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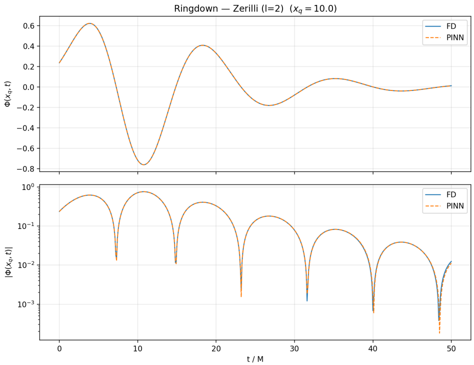
The physical picture — black-hole ringdown

A perturbed Schwarzschild black hole “rings” like a struck bell: after the merger, the geometry settles down by emitting **damped sinusoidal** gravitational waves — the **Quasi-Normal Modes (QNMs)**.

$$h(t) \;\propto\; e^{-t/\tau}\cos(\omega\, t + \phi_0)$$

- ω — ringing frequency (real part).
- $\tau = -1/\Im\omega$ — damping time (imaginary part).
- Determined entirely by the black hole's mass M and spin a.
- Observed by LIGO/Virgo in every binary-black hole detection since GW150914.



Ringdown waveform $\Phi(x_q = 10M,\ t)$

From Einstein's equations to a 1+1D wave problem

Step 1 — background. Schwarzschild geometry (M is the black hole mass),

$$\mathrm{d}s^2 \;=\; -f(r)\,\mathrm{d}t^2 + f(r)^{-1}\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \tfrac{2M}{r}.$$

Step 2 — linear perturbation. Write $g_{\mu\nu} = g^{(0)}_{\mu\nu} + h_{\mu\nu}$ with $|h| \ll |g^{(0)}|$ and expand the vacuum Einstein equations to first order: $\delta R_{\mu\nu}[h] = 0$.
Step 3 — exploit spherical symmetry. Decompose $h_{\mu\nu}$ in *tensor spherical harmonics*; the modes split into two **decoupled parity sectors**:

Sector	Parity	Master equation
Axial (odd)	$(-1)^{\ell+1}$	Regge–Wheeler
Polar (even)	$(-1)^{\ell}$	Zerilli

Each sector reduces to a single scalar field $\Phi(t, r_*)$ on the **tortoise coordinate** $\mathrm{d}r_* = \mathrm{d}r/f(r)$.

The two master equations — Regge–Wheeler and Zerilli

Both sectors obey the **same 1+1D wave form**

$$-\frac{\partial^2 \Phi}{\partial t^2} + \frac{\partial^2 \Phi}{\partial {r_*}^2} - V_\ell(r)\,\Phi = 0,$$

but with two *different* effective potentials:

Regge–Wheeler (axial)

$$V_\ell^{\mathrm{RW}}(r) = f(r)\left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3}\right]$$

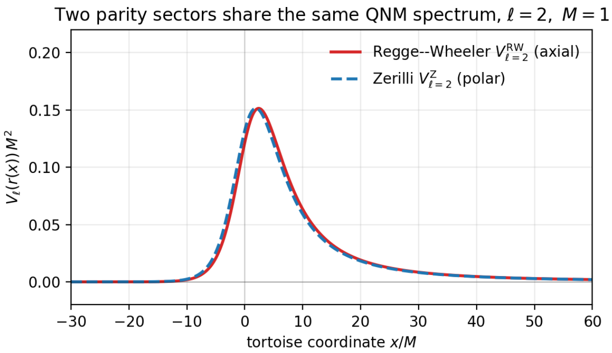
Zerilli (polar)

$$V_\ell^{\mathrm{Z}}(r) = f(r)\,\frac{2\big[n^2(n+1)r^3 + 3n^2 M r^2 + 9nM^2 r + 9M^3}{r^3(nr+3M)^2}$$

$$n = \tfrac{1}{2}(\ell-1)(\ell+2).$$

Two *different* potentials, one per parity sector — but they encode *the same physics*, as the next slide shows.

Isospectrality — one spectrum, two potentials



Chandrasekhar–Detweiler isospectrality: the two potentials are *distinct* but share *exactly the same* QNM spectrum $\{\omega_n\}$.

$\Rightarrow$ Either equation is a valid target for our solver.

This work uses the **Zerilli (polar, $\ell = 2$)** form, matching the reference PINN literature (Patel *et al.* 2024) and giving a single, well-defined benchmark to push accuracy against.

The numerical problem we actually solve

Target: the polar $\ell=2$ Zerilli equation on $r_*/M \in [-50, 150]$, $t/M \in [0, 50]$.

- **Initial condition** ($t = 0$): localised Gaussian pulse.
- **Boundary conditions:** *purely outgoing* at $r_* \to +\infty$, *purely ingoing* at $r_* \to -\infty$ (horizon).
- These BCs are precisely what selects the discrete **QNM spectrum** $\omega_n \in \mathbb{C}$.

Reference fundamental mode ($\ell=2$, $n=0$, Leaver 1985):

$$M\omega \;=\; 0.3737 - 0.0890\, i, \qquad \tau/M \;=\; -1/\Im\omega \;=\; 11.241.$$

Every result in this talk is benchmarked against this number.

Three ways to compute QNMs

1. Leaver continued fraction (1985) — the *exact reference*

Frobenius ansatz $\Rightarrow$ 3-term recurrence $\Rightarrow$ continued fraction. **+** 12-digit accuracy in ms. **–** only known analytic potentials.

2. Finite-difference (FD) time evolution — the *ground-truth waveform*

2nd-order centred FD on the (r_*, t) grid; extract ω, τ from the late-time signal. **+** full (x, t) field. **–** *expensive when accurate*.

3. Neural solvers — *our tools*

(a) PINN: mesh-free, differentiable solver for *one* configuration.
(b) Hybrid FNO: a cheap coarse FD solve + a neural correction, trained *once* over a *family* of black holes.

Why do we want a cheap, generalising solver?

QNM extraction needs the **time-domain waveform** — i.e. a PDE solve. Classical FD is accurate but its cost *explodes* exactly where the science is:

- **High oscillation.** Higher multipoles ℓ and higher spin push the ringdown to *shorter wavelengths*; resolving them forces a finer grid, and FD cost scales as $\sim k^{d+1}$ in the refinement factor k.
- **High dimension.** Schwarzschild is 1+1D; Kerr (Teukolsky) is 2+1D. Each added dimension multiplies the grid cost.
- **Many-query regimes.** Parameter estimation, population studies and surrogate building need *thousands* of solves over $(M, a, \text{initial data})$.

The two neural answers in this talk

PINN — a *differentiable* mesh-free solver: a single trained network, but *one solve per configuration*.

Hybrid FNO — keep a *cheap* coarse solver as the workhorse and learn only the part it cannot resolve, *amortised* over a whole parameter family with a single forward pass per new black hole.

What this project contributes

Schwarzschild $\ell = 2$ is the **stress test** for QNM surrogates — every error is measurable against an exact reference.

1. **Forward PINN, accuracy push:** hard BCs, gPINN residuals, **greedy adaptive sampling**, extended L-BFGS. Field error to **0.58%** (Patel: 28%, a $\sim 50\times$ reduction) and ω to **0.03%** (M4).
2. **New extraction methodology — M1–M5:** from FFT/NLS to ESPRIT and *plateau scans* that give principled, falsifiable (ω, τ) with systematic-error bars.
3. **Hybrid coarse-FD + neural-residual surrogate:** a *label-free*, *gated* FNO that super-resolves a cheap $k = 4$ coarse solve by $\sim 10\times$ in field error, generalising across a 1000-black hole family at $1/64$ of the fine-FD cost.

This talk presents our **best forward PINN** and the **hybrid FNO**, and compares both against the reference paper.

Forward PINN — problem statement

Goal. Train a network $\Phi_{\boldsymbol{\theta}}(x,t)$ to satisfy the Zerilli PDE on the full (x,t) domain, plus its initial and boundary conditions.

We minimise a sum of **seven** weighted mean-squared-error terms,

$$\mathcal{L}_{\text{fwd}}(\boldsymbol{\theta}) = \underbrace{\lambda_r\mathcal{L}_r + \lambda_{r_x}\mathcal{L}_{r_x} + \lambda_{r_t}\mathcal{L}_{r_t}}_{\text{PDE residual + gPINN}} + \underbrace{\lambda_{ic}\mathcal{L}_{ic} + \lambda_{iv}\mathcal{L}_{iv}}_{\text{initial condition}} + \underbrace{\lambda_{bl}\mathcal{L}_{bl} + \lambda_{br}\mathcal{L}_{br}}_{\text{boundary conditions}}$$

Each $\mathcal{L}_i$ is a mean of squared residuals over a sample of collocation points; the next two slides

write the seven sums explicitly.

Forward PINN loss — PDE residual + gPINN

PDE residual of the Zerilli equation,

$$r_\theta(x,t) \equiv -\,\partial_t^2\Phi_\theta + \partial_x^2\Phi_\theta - V_\ell(r)\,\Phi_\theta.$$

At $N_r = 32{,}000$ collocation points we evaluate

$$\mathcal{L}_r = \tfrac{1}{N_r}\sum_i \big[r_\theta(x_i,t_i)\big]^2 \quad \text{(standard PINN residual)},$$

$$\mathcal{L}_{r_x} = \tfrac{1}{N_r}\sum_i \big[\partial_x r_\theta\big]^2, \qquad \mathcal{L}_{r_t} = \tfrac{1}{N_r}\sum_i \big[\partial_t r_\theta\big]^2.$$

The two extra terms are the **gPINN** regularisers (Yu *et al.* 2022): penalising the *gradient* of the residual forces it to be smoothly small, not just point-wise small, and gives noticeably cleaner derivatives.

Forward PINN loss — initial and boundary conditions

Initial condition at $t = 0$ on N_{ic} spatial points:

$$\mathcal{L}_{\mathrm{ic}} = \frac{1}{N_{\mathrm{ic}}} \sum_j \left[\Phi_\theta(x_j, 0) - \Phi_0(x_j)\right]^2 \quad (\Phi_0 \text{ a Gaussian pulse}),$$

$$\mathcal{L}_{\mathrm{iv}} = \frac{1}{N_{\mathrm{ic}}} \sum_j \left[\partial_t \Phi_\theta(x_j, 0)\right]^2 \quad \text{(time-symmetric data)}.$$

Boundary conditions on N_{bc} time points — outgoing at x_{max}, ingoing at x_{min}:

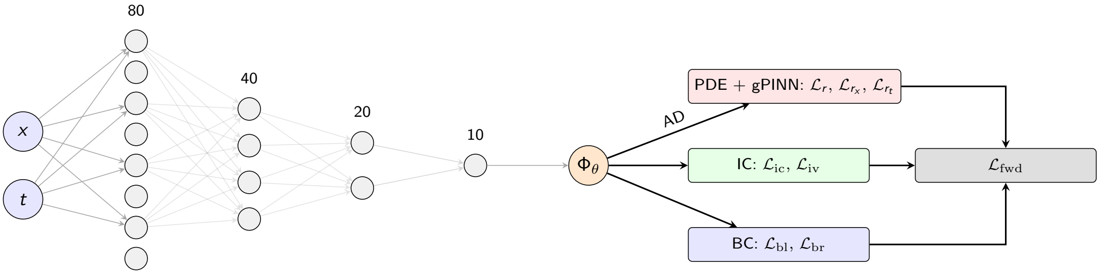
$$\mathcal{L}_{\mathrm{bl}} = \frac{1}{N_{\mathrm{bc}}} \sum_k \left[(\partial_t - \partial_x)\Phi_\theta(x_{\mathrm{min}}, t_k)\right]^2,$$

$$\mathcal{L}_{\mathrm{br}} = \frac{1}{N_{\mathrm{bc}}} \sum_k \left[(\partial_t + \partial_x)\Phi_\theta(x_{\mathrm{max}}, t_k)\right]^2.$$

Forward PINN — training setup

- **Network.** Fully-connected MLP, widths $[\,2, 80, 40, 20, 10, 1\,]$, tanh ($\sim$4,500 params).
- **Output transform.** $\Phi_\theta = A \tanh(\widetilde{\Phi}_\theta)$, $A = 1$ — bounds the field, stabilises early training.
- **Loss weights.** $\boldsymbol{\lambda} = (100, 100, 100,\ 1, 100,\ 1, 1)$.
- **Optimiser.** Adam for 10,000 steps to find a basin, then **L-BFGS** for 30,000 steps to converge inside it.
- **Sampling.** Collocation points re-drawn every epoch via *greedy resampling* (next three slides) — our key training innovation.

Forward PINN — architecture

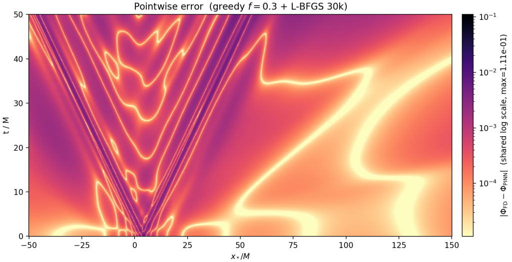


Inputs (x, t) feed a four-layer tanh MLP ($80 \to 40 \to 20 \to 10$, $\sim$4,500 params); the output passes through $\Phi_\theta = A \tanh(\widetilde{\Phi}_\theta)$ and is differentiated by automatic differentiation (**AD**) to form the seven residuals.

Innovation — why adaptive sampling matters

Uniform collocation wastes points where the residual is already small and starves the network where it is large.

- **Residual-Adaptive Distribution (RAD)** [Wu *et al.* 2023]: probabilistic resampling $p_i \propto |r_i|^k + c$. Two hyperparameters.
- Stochastic $\Rightarrow$ extreme residuals not always selected.
- Sampling noise can interfere with Adam dynamics.



Final error map: residual concentrates along the wavefronts travelling out from the burst.

Innovation — greedy resampling algorithm

Replace probabilistic RAD with **deterministic top-N** selection:

1. Every P Adam steps, generate $N_{\text{cand}} = 100{,}000$ random candidates in the domain.
2. Evaluate the PDE residual $|r_i|$ at each (one batched forward pass).
3. **Sort by** $|r_i|$ **descending**; keep the top $f \cdot N_r$.
4. Fill the remaining $(1-f)N_r$ slots with **uniform random** points (preserves exploration).
5. Replace all $N_r = 32{,}000$ collocation points; continue Adam.

Advantages over RAD: *deterministic* (worst residuals always kept); a *single* hyperparameter f (0 = uniform, 1 = fully greedy); no $|r|^k + c$ shape to tune.

Innovation — greedy f sweep

Configuration	**RMSD**	**vs. base**
Uniform (baseline)	0.00183	—
RAD $k = 2$	0.00095	-48%
Greedy $f = 0.3$	**0.00085**	-54%
Greedy $f = 0.5$	0.00107	-42%
Greedy $f = 0.8$	0.00165	-10%
Greedy $f = 0.3$ + L-BFGS 30k	**0.00082**	-55%

- $f = 0.3$ **optimal:** 30% greedy + 70% uniform balances exploitation and exploration.
- Too aggressive ($f \geq 0.5$): network over-fits the high-residual band and regresses elsewhere.
- **Beats RAD with a simpler, deterministic, one-knob algorithm.**
- Combined with extended L-BFGS $\Rightarrow$ our best forward PINN.

QNM Extraction Methods

Every solver (PINN, hybrid FNO, FD) outputs a waveform $\Phi(x_q, t)$; the QNM is then *extracted*. We report **five** methods and quote the best per quantity.

- **M1 — FFT + log-linear envelope:** ω from the FFT peak, τ from the $\log|\text{peaks}|$ slope. Fast, but single-window.
- **M2 — single-mode NLS:** fit $Ae^{-t/\tau}\cos(\omega t + \phi)$ over a window. Best wide-window τ estimator; biased by overtones early.
- **M3 — ESPRIT**, **M4 — 1-D plateau scan**, **M5 — 2-D plateau scan** are detailed on the next slides.

Two problems M1/M2 alone cannot solve: *overtone contamination* ($\tau_1 \approx 3.65M$ near the burst) and *window dependence* (a single number with no sensitivity $\Rightarrow$ not falsifiable).

M3 — ESPRIT subspace decomposition

Idea: model the uniformly-sampled ringdown as a sum of K complex exponentials (poles) and recover them *simultaneously* by a subspace rotation — no nonlinear fit and no initial guess (Estimation of Signal Parameters via Rotational Invariance Techniques).

$$\Phi(x_q,\, k\Delta t) \;\approx\; \sum_{n=0}^{K-1} c_n\, z_n^{\,k}, \qquad z_n = e^{(-1/\tau_n + i\omega_n)\,\Delta t}.$$

Procedure:

1. Hilbert-transform the real signal to its analytic form, so each physical mode appears once.
2. Build the Hankel data matrix from the samples; SVD $\to$ signal subspace $\mathbf{U}$.
3. Solve the rotational-invariance eigenproblem $\mathbf{U}_1^{\dagger}\mathbf{U}_2 = \boldsymbol{\Psi}$; the poles are $z_n = \mathrm{eig}(\boldsymbol{\Psi})$.
4. Read off $\omega_n = \Im(\log z_n)/\Delta t,\;\; \tau_n = -\Delta t/\Re(\log z_n)$; report the largest-amplitude mode.

Strength: no initial guess; resolves multiple modes naturally. **Weakness:** Hankel conditioning degrades on a clean single-mode tail, so on our QNM-dominated fields ESPRIT can return spurious machine-noise poles.

M4 — two-mode NLS with start-time plateau scan

Step 1. Fit a *two-mode* template (fundamental + first overtone):

$$\Phi_{\text{model}}(t) = A_0 e^{-t/\tau_0} \cos(\omega_0 t + \phi_0) + A_1 e^{-t/\tau_1} \cos(\omega_1 t + \phi_1)$$

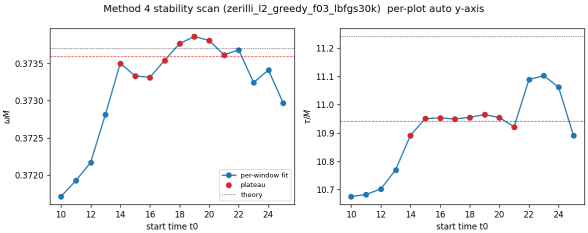
on $[t_0, t_{\text{max}}]$; identify the longer-τ mode as the fundamental.

Step 2. Repeat over start times $t_0 \in [10, 25]M$; find the contiguous **plateau** of low-scatter consecutive fits.

Step 3. Report $\omega_{M4} = \langle\omega\rangle_{\text{plateau}}$, $\sigma_\omega = \mathrm{std}(\omega)_{\text{plateau}}$ — the plateau spread is a **principled systematic error bar**.

An honest extraction must be invariant to small window shifts; the plateau is the empirical signature of that invariance.

M4 — example stability scan



Forward PINN: M4 stability scan over $t_0 \in [10, 25]M$.

- ω (top): flat across $t_0 \in [14, 21]M \Rightarrow$ plateau.
- τ (bottom): stabilises once t_0 is past the prompt burst.
- Plateau quote $\omega_{M4} = 0.37359 \pm 1.96 \times 10^{-4}$ $\Rightarrow$ **0.029%** error.
- Green band $= 2\sigma_\omega$: an honest, falsifiable systematic.

M5 — two-mode NLS with 2-D plateau scan

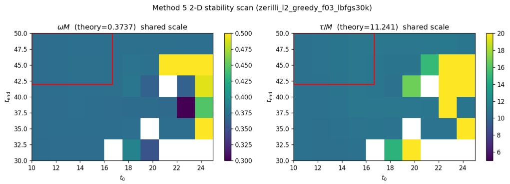
M4 sweeps only one window edge (t_0). **M5** sweeps both:

$$[t_0, t_{\text{end}}] \in [10, 25]M \times [t_{\min}, 50]M \quad \text{on a } 10\times 6 \text{ grid},$$

reporting the mean/stddev over the rectangular plateau of smallest joint scatter.

- t_0 scans control *overtone leakage* (early edge).
- t_{end} scans control *Price-tail leakage* (late edge).
- Truncating either edge prematurely is a *model-misspecification* bias — it does *not* vanish with more data.
- To our knowledge the **first systematic 2-D plateau analysis** for neural-solver ringdowns.

M5 — example 2-D heatmap



Top: ω map. **Bottom:** τ map.

- Axes: t_0 (start) vs t_{end} (end).
- Colour: local relative scatter (capped 5%).
- **Red box:** plateau (minimum scatter).
- Grey: NaN cells where the two-mode fit failed.

Plateau quote: $\omega_{M5} = 0.3732$ (0.13%), $\tau_{M5} = 10.82$ (3.91%). M5 reproduces M4 *and* certifies the right-edge choice is harmless.

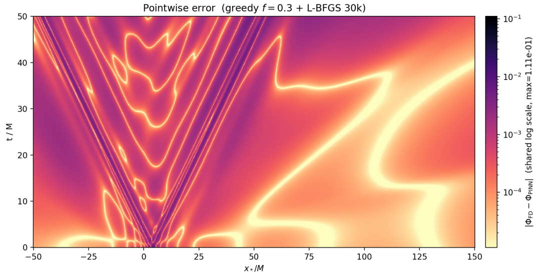
Reporting convention — best estimator per quantity

Different quantities are best estimated from different features of the waveform, so we quote a **best-per-quantity headline** (and list all five as cross-checks):

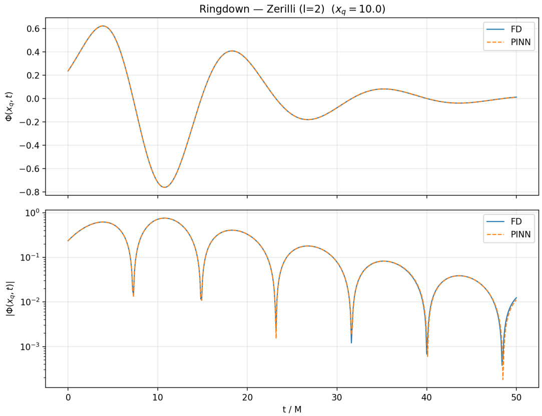
Quantity	Estimator	Reason
ω	**M4 / M5 plateau**	phase aggregated across cycles; lowest variance.
τ	**M2 wide window** or **M4/M5 plateau**	amplitude over the longest clean decay range.

- Consistent with NR ringdown practice (Berti *et al.* 2007).
- Patel *et al.* 2024 use a similar split but report *no* fit-window sensitivity.
- **From here on, every result quotes field error + the best M1–M5 fit for ω and τ, naming the extractor.**

Forward PINN — waveform agreement

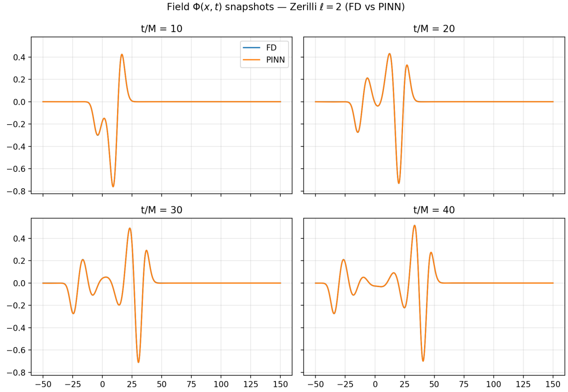


Pointwise error $|\Phi_{PINN} - \Phi_{FD}|$ over the full domain.

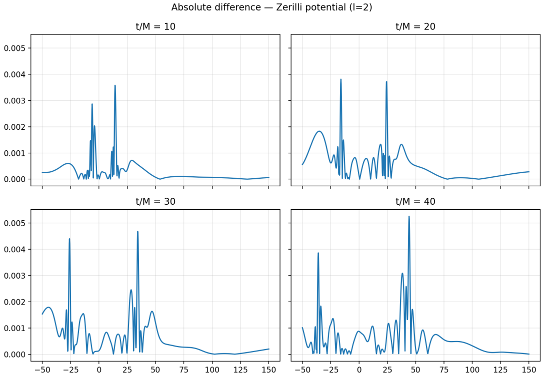


Ringdown overlay at $x_q = 10M$: PINN (solid) vs FD (dashed). Greedy $f = 0.3$ + L-BFGS 30k.

Forward PINN — snapshots and pointwise error



Φ_{PINN} (solid) vs Φ_{FD} (dashed) at $t/M \in \{10, 20, 30, 40\}$.



Pointwise $|\Phi_{PINN} - \Phi_{FD}|$: residual $\lesssim 5 \times 10^{-3}$ near the barrier, $\lesssim 10^{-3}$ far away.

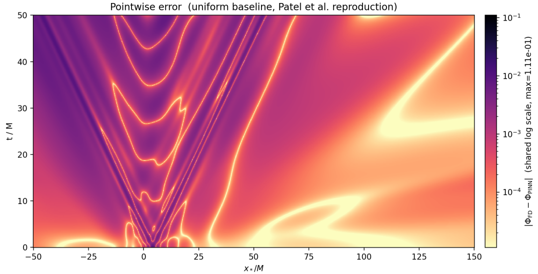
Forward PINN — best result: field + QNM

Field accuracy. Gridwise RMSD vs FD $= 8.17 \times 10^{-4}$ (relative $L^2 \approx 0.58\%$).

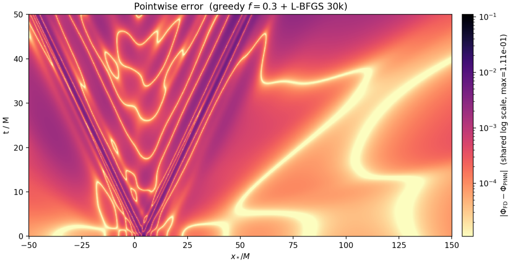
Method	$\boldsymbol{\omega}$	$\boldsymbol{\omega}$ **err**	$\boldsymbol{\tau}$	$\boldsymbol{\tau}$ **err**
Truth (Leaver)	0.3737	—	11.241	—
M1 (FFT + envelope)	0.3712	0.68%	10.92	2.83%
M2 (single-mode NLS)	0.3794	1.54%	11.46	**1.99%**
M3 (ESPRIT)	0.4829	29.2%	7.84	30.3%
M4 (two-mode plateau)	0.37359	**0.029%**	10.94	2.65%
M5 (2-D plateau)	0.37284	0.23%	10.80	3.91%

- **Best ω: M4 at 0.029%** — plateau + two-mode template removes overtone bias *and* window arbitrariness.
- **Best τ: M2 wide-window at 1.99%.**
- M3 fails here (Hankel ill-conditioned at $\sim 10^{-24}$ late amplitude) — flagged automatically; the five-method spread is the cross-check.

Forward PINN — greedy vs uniform error map



Uniform baseline (Patel *et al.* 2024 reproduction): residual along the outgoing wavefronts.



Greedy $f = 0.3$ + L-BFGS 30k (same colour scale): wavefront residual visibly reduced (overall RMSD −55% vs uniform).

Patel *et al.* 2024: field error 28%, ω error 0.29%. **Our forward PINN: field** 0.58% **(**$\sim$50$\times$ **tighter),** ω 0.03% **(M4).**

Primer — what is a Fourier Neural Operator?

A PINN learns *one* solution function $\Phi(x, t)$ and must be retrained for every new problem. A **neural operator** instead learns the *solution map* itself — a mapping between whole functions:

$$\mathcal{G}_\theta : \; a(x) \;\longmapsto\; u(x),$$

e.g. (initial data / coefficients) $\mapsto$ (solution field). Once trained it returns the solution for *any* new input a in a single forward pass — no re-solving.

Architecture: lift $\to$ Fourier layers $\to$ project

$$a(x) \xrightarrow{P} v_0 \xrightarrow{\text{Fourier}} v_1 \to \cdots \to v_T \xrightarrow{Q} u(x)$$

- **Lift** P: pointwise (1×1) MLP raises the few physical channels to a wide hidden width d_v.
- T **Fourier layers** (here $L=4$): the global mixing — next slide.
- **Project** Q: pointwise MLP maps the hidden width back to the output field.

Key property: the learnable weights live in *Fourier space*, not on grid nodes $\Rightarrow$ the operator is (approximately) **discretisation-invariant** — train on one grid, evaluate on a finer one.

Primer — inside a Fourier layer (the spectral convolution)

Each layer updates the hidden function with a *local* path plus a *global* path:

$$v_{t+1}(x) = \sigma\Big(\underbrace{W\, v_t(x)}_{\text{local } 1\times 1 \text{ conv}} \;+\; \underbrace{(\mathcal{K} v_t)(x)}_{\text{global spectral term}} \Big).$$

The global term is a convolution, evaluated cheaply in frequency space (convolution theorem): FFT $\to$ keep the lowest $k_{\max}$ modes $\to$ multiply each by a learned complex weight $\to$ inverse FFT,

$$(\mathcal{K} v_t)(x) = \mathcal{F}^{-1}\big(R_\theta \cdot \mathcal{F}(v_t)\big)(x), \qquad R_\theta \in \mathbb{C}^{\,k_{\max}\times d_v \times d_v}.$$

- **Global receptive field in one layer** — every point couples to every other, matching the globally-coupled nature of PDE solution operators.
- **Mode truncation** (modes $> k_{\max}$ dropped): a low-frequency / smoothness prior, and cheap — $\mathcal{O}(N \log N)$ via the FFT, not $\mathcal{O}(N^2)$.
- **Resolution invariance:** R_θ is indexed by mode number, not grid index — refine the grid and the *same* weights still apply.
- The local W path restores the sharp / non-periodic features mode truncation discards.

Next: we do not learn the *whole* wave operator with an FNO — we use one only for the bounded *residual* a

Hybrid FNO — the idea

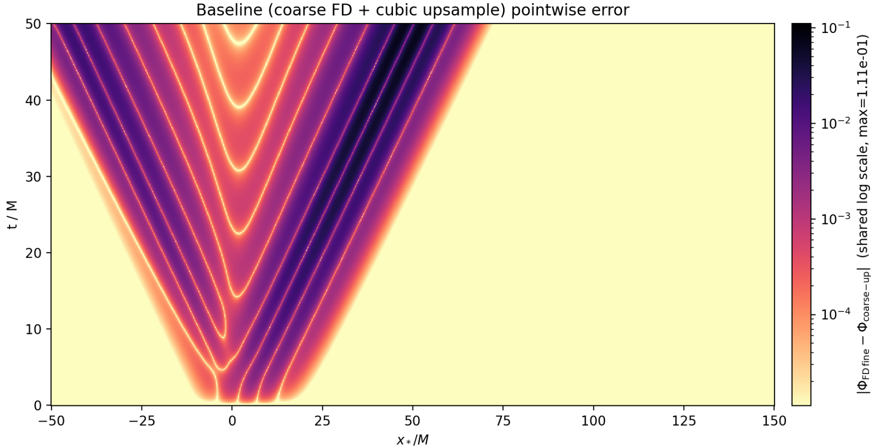
A PINN (or an end-to-end neural operator) learns the *entire* wave-propagation operator from scratch. **Alternative:** keep a cheap classical solver as the workhorse and ask the network *only* for the part it cannot resolve.

Hybrid coarse-FD + neural-residual surrogate

1. Run *one* inexpensive **coarse** ($k=4$) FD solve; upsample it to the fine grid $\Rightarrow$ the *prior* $\mathcal{U}\Phi^{c}$.
2. A 2-D FNO predicts the **discretisation residual** the coarse solve misses.
3. The target is built by **label-free Richardson extrapolation** of two coarse solves — *no fine-grid ground truth is ever used in training*.
4. A parameter-free **gradient gate** confines the correction to the under-resolved wavefronts.

Cost of the classical part at inference: $k^{-3} = 1/64$ of a fine FD solve.

Hybrid FNO — where the coarse prior fails



Pointwise error of the cheap $k = 4$ coarse prior against the fine-FD reference (same magma_r log scale as every error map in this talk). The error is **concentrated on the outgoing wavefronts** and near-zero in the smooth interior and causal exterior — a *bounded, spatially-localised* target. This is exactly the structure the gated hybrid FNO is built to repair.

The full hybrid surrogate pipeline

See report, p. XX (Sec. 3.6, Fig. 4 — hybrid surrogate pipeline).

At a glance: coarse $k=4$ FD $\rightarrow$ quintic upsample (prior) $\rightarrow$ 2-D FNO residual $\rightarrow$ gradient gate $\rightarrow$ + prior $= \Phi^{\mathrm{hyb}}$.

Training target (label-free): Richardson extrapolation of the $k=4$ and $k=2$ coarse solves.

Hybrid FNO — label-free Richardson target

The $k = 4$ prior is deliberately weak: its default 2nd-order stencil leaves an $\mathcal{O}(\Delta x_{\mathrm{c}}^2)$ dispersion error (6.1% relative L^2). Rather than buy a better stencil, we **cancel** the leading error with two coarse solves:

$$\Phi^{\mathrm{R}} = \frac{4\,\mathcal{U}\Phi_2^{\mathrm{c}} - \mathcal{U}\Phi^{\mathrm{c}}}{3} = \Phi + \mathcal{O}(\Delta x_{\mathrm{c}}^4), \qquad \delta\Phi_{\mathrm{R}} = \Phi^{\mathrm{R}} - \mathcal{U}\Phi^{\mathrm{c}}.$$

- The network is trained to predict $\delta\Phi_{\mathrm{R}}$ — its loss **never sees the fine FD field**. Self-supervised from coarse solves.
- At inference only the single $k = 4$ solve is needed ($k^{-3} = 1/64$ of fine-FD cost); the $k = 2$ solve is a training artefact.
- Quintic upsample $\mathcal{U}$: $\mathcal{O}(\Delta x_{\mathrm{c}}^6)$ interpolation error stays below the $\mathcal{O}(\Delta x_{\mathrm{c}}^4)$ Richardson signal.

Weak prior + **Richardson** is the opposite of a strong classical stencil: cheapest possible solver, network does correspondingly more — the regime where the k^{d+1} saving is largest and which scales to higher dimensions.

Hybrid FNO — the gradient gate (key innovation)

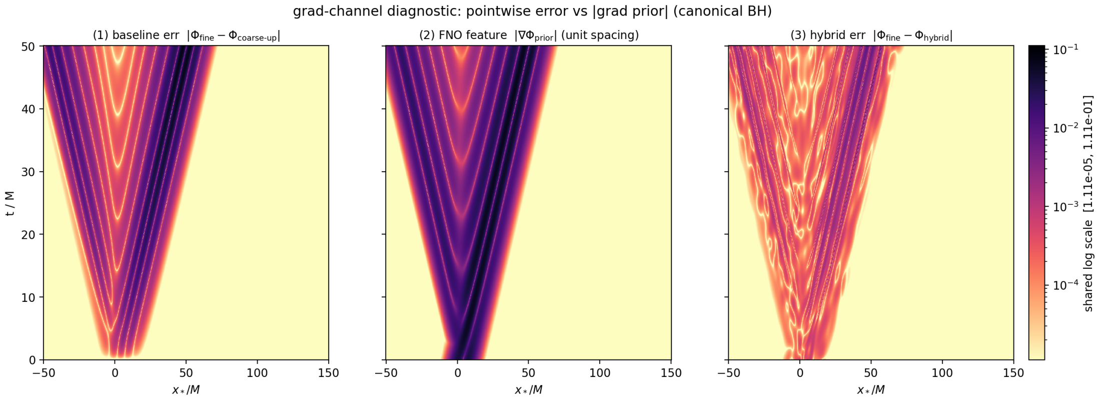
The network correction is modulated by a **parameter-free** gate before being added to the prior:

$$\Phi^{\mathrm{hyb}} = \mathcal{U}\Phi^{\mathrm{c}} + g(x,t)\,\mathcal{H}_\theta(X), \qquad g = \frac{|\nabla \mathcal{U}\Phi^{\mathrm{c}}|}{|\nabla \mathcal{U}\Phi^{\mathrm{c}}| + s}, \quad s = 10^{-3}.$$

- $g \to 1$ on the high-gradient **wavefronts**, where the coarse prior dephases and carries its dispersion error.
- $g \to 0$ in the smooth interior and the causal exterior, where the prior is already machine-accurate.
- Adds *no* trainable parameters; the scale s is fixed *label-free* from the prior gradient distribution (68th percentile), within 2% of the field-RMSD optimum.
- **Hard guarantee:** wherever $g = 0$ the hybrid *is* the coarse prior — the network cannot pollute the regions the solver already resolves.

The gate rests on a testable hypothesis — *the coarse-prior error is large exactly where the prior gradient is large* — which the next slide verifies.

The gate hypothesis, verified



Grad-channel diagnostic on the canonical BH, all panels on the shared log scale: **(1)** coarse-prior error $|\Phi^{\rm f} - \mathcal{U}\Phi^{\rm c}|$, **(2)** the prior-gradient feature $|\nabla\mathcal{U}\Phi^{\rm c}|$ that defines the gate, **(3)** hybrid residual $|\Phi^{\rm f} - \Phi^{\rm hyb}|$. Panels (1) and (2) share the *same* wavefront structure — the prior fails exactly where its gradient is large (per-cell log–log Pearson $r = 0.917$) — so the gate routes the correction there; panel (3) shows it repaired.

Hybrid FNO — network and cost

Residual operator $\mathcal{H}_\theta$ (2-D FNO):

- **5-channel input:** upsampled prior, $\Phi^c(x, 0)$, potential $V_\ell(x; M)$, mass M, coarse initial velocity Π^c_0.
- lift 1×1 conv $5\to32$; $L=4$ Fourier layers; projection $32\to1$.
- $\sim 1.13 \times 10^6$ parameters ($\sim 70\times$ smaller than the end-to-end FNO).
- Adam, lr 10^{-3}, label-free target, fine grid 1001×501.

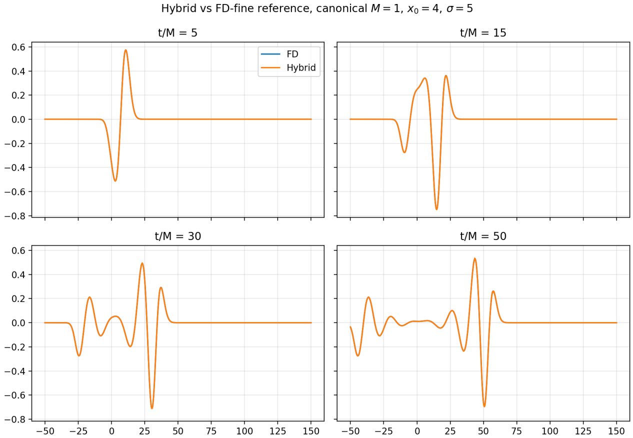
Why hybrid, not end-to-end?

The network sits *in series* with a classical solver that already gives a good guess, so it **inherits the stability and discretisation-invariance** of the FD scheme as a hard prior, and only repairs a bounded, localised residual.

Deployment cost

One $k=4$ solve (1/64 fine-FD) + one FNO forward pass, for *any* (M, x_0, σ) in the family — *no retraining*.

Hybrid FNO — field accuracy



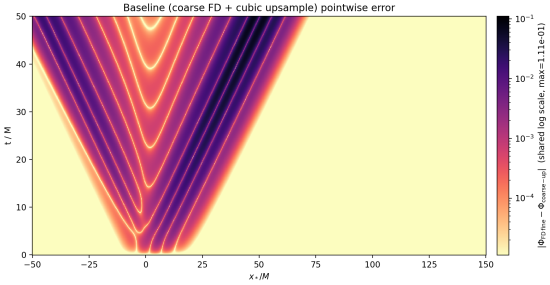
Hybrid (orange) vs fine FD (blue) at $t/M \in \{5, 15, 30, 50\}$ — indistinguishable on this scale.

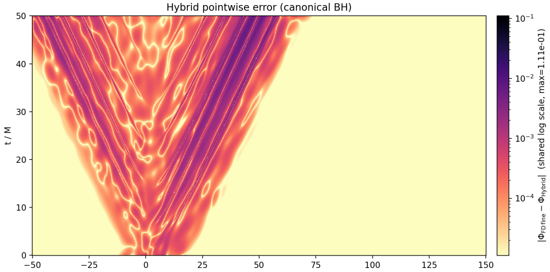
Field error (relative L^2):

	coarse prior	**hybrid**
canonical black hole	4.93%	0.49%
100-black hole median	6.11%	0.67%

- **10.1**× field gain on the canonical black hole, **9.1**× across the population (79× in MSE).
- Consistent across the whole training family — not a single-draw artefact.
- The residual lives *only* on the outgoing wavefronts.

Hybrid FNO — the gate works





Error stratified by prior gradient (canonical black hole):

decile	baseline	ratio
90–100% (wavefronts)	1.3×10^{-2}	**12.3**$\times$
80–90%	4.4×10^{-3}	$9.2\times$
70–80%	2.2×10^{-3}	$7.2\times$
$<$ median	$\sim$floor	$\sim1\times$

- The correction tracks the gate: largest on wavefronts, vanishing in already-resolved regions.
- Over the 60.5% of the domain the prior already nails, the gate holds network speckle to $\sim10^{-6}$.

Hybrid FNO — QNM extraction (median over 100 BHs)

	Hybrid		Coarse prior		Fine FD	
Method	$\lvert\Delta\omega\rvert$%	$\lvert\Delta\tau\rvert$%	$\lvert\Delta\omega\rvert$%	$\lvert\Delta\tau\rvert$%	$\lvert\Delta\omega\rvert$%	$\lvert\Delta\tau\rvert$%
M1 (FFT+log)	0.511	0.394	0.441	0.324	0.485	0.262
M2 (1-mode NLS)	0.422	0.818	0.396	0.860	0.436	0.805
M3 (ESPRIT)	0.748	1.855	0.766	1.878	0.787	1.907
M4 (1-D plateau)	**0.049**	0.263	0.051	0.174	0.014	0.063
M5 (2-D plateau)	0.052	0.254	0.051	0.180	0.014	0.059

- **Best ω: M4 at 0.049%**; on the canonical black hole, M4 gives ω **0.045%**, τ **0.083%**.
- **Honest finding:** at a single observer the coarse prior is already at the extractor floor — the hybrid is *tied* with it on the QNM, neither improving nor degrading it.
- **The hybrid's demonstrated value is the global field ($\sim 10\times$) and generalisation**, not a single-observer QNM the cheap prior already resolves.

Patel 2024 vs. our PINN vs. our Hybrid FNO

All at the *identical* configuration ($\ell = 2$ Zerilli, $M = 1$, $x_0 = 4$, $\sigma = 5$). Field error is the relative L^2 vs the FD reference; ω, τ are best-of-M1–M5.

Model	**field err (rel-L^2)**	**best ω**	**best τ**	**generalises?**
Patel *et al.* 2024 (PINN)	28.06%	0.29% (M1)	10.25% (M1)	no
Our forward PINN	0.58%	**0.029%** (M4)	1.99% (M2)	no
Our hybrid FNO	0.49%	0.045% (M4)	**0.083%** (M4)	**yes**
coarse prior (no NN)	4.93%	0.051% (M4)	0.174% (M4)	—
fine FD (truth)	—	0.014% (M4)	0.063% (M4)	—

- **Both our models crush Patel:** field $\sim 50\times$, ω $\sim 10\times$, τ by 5–120$\times$.
- τ **is the discriminating quantity —** everyone does fine on ω; our plateau pipeline pulls τ from $\sim 10\%$ to $\sim 0.1\%$.
- **Honest:** the single-observer QNM sits at the prior + extractor floor (hybrid $\approx$ coarse prior); the hybrid's unique win is field super-resolution + amortised generalisation.

Where each model wins

Capability	Best model	Why
Single-config ω accuracy	forward PINN	0.029% via M4 plateau
Damping time τ	hybrid / PINN	$\sim$0.08% (M4), $\sim$120$\times$ vs Patel
Field super-resolution	**hybrid FNO**	10$\times$ over a cheap coarse solve
Generalise over a family	**hybrid FNO**	one model, 1/64 cost, no retraining
QNM from coarse grid	coarse FD already	grid-robust eigenvalue

The take-home

The ringdown *eigenvalue* is robust and cheap to get; the fine *field* structure is what needs machine learning. The hybrid FNO buys **amortised, generalising field super-resolution** a single-instance PINN structurally cannot — at a fraction of the classical cost.

Towards Kerr — where the hybrid should pay off

Schwarzschild was the controlled benchmark. The hybrid is built for the regime where classical solves are *expensive*:

- **Kerr (spinning) black holes:** the **Teukolsky** equation, 2+1D — one extra dimension multiplies grid cost, exactly where a $1/k^{d+1}$ coarse prior saves most.
- **Spin-graded difficulty:** higher spin $\Rightarrow$ tighter light ring $\Rightarrow$ shorter QNM wavelength $\Rightarrow$ a fixed coarse grid under-resolves *more*. We expect the network's correction — and its headroom — to *grow with spin*.
- **Higher multipoles ($\ell = 4$):** beyond the $\ell = 2$ that the PINN-for-QNM literature (incl. Patel *et al.*) demonstrates.

Status

Kerr corpus (mode-selective QNM extraction, label-free Richardson target on a spin-graded coarse ladder) is *in progress*; first $\ell = 4$ both-axes (field + QNM) runs are queued.

Summary

1. **Forward PINN:** hard BCs, gPINN, **greedy adaptive sampling**, extended L-BFGS $\Rightarrow$ field rel-L^2 0.58% ($\sim$50$\times$ below Patel's 28%), ω to **0.029%** (M4).
2. **Extractors M1–M5:** plateau scans give principled, falsifiable (ω, τ) with systematic-error bars; τ to $\sim$0.1%.
3. **Hybrid FNO:** label-free, gated coarse-FD $+$ neural-residual surrogate $\Rightarrow$ 10$\times$ **field super-resolution** at 1/64 cost, generalising across a 1000-black hole family; QNM preserved at the extractor floor.
4. **Next:** Kerr / Teukolsky, where the cheap-prior saving and the spin-graded headroom are largest.

Thank you

Questions?

Appendix — Richardson extrapolation: derivation

Setup. A p-th order scheme on grid spacing h admits an asymptotic error expansion

$$\Phi^{\mathrm{c}}(h) \;=\; \Phi \;+\; C\,h^{p} \;+\; D\,h^{p+2} \;+\; \mathcal{O}(h^{p+4}),$$

with Φ the exact solution and C, D independent of h (functions of the solution's derivatives). Our coarse solver is the centred 2nd-order method-of-lines scheme, so $p = 2$ and — because centred stencils have a *symmetric* truncation error — the expansion runs in *even* powers of h.

Two coarse solves at spacings h and $h/2$ (here the $k\!=\!4$ and $k\!=\!2$ grids, ratio 2):

$$\Phi^{\mathrm{c}}(h) = \Phi + Ch^2 + Dh^4 + \mathcal{O}(h^6), \qquad \Phi^{\mathrm{c}}(h/2) = \Phi + \tfrac{C}{4}h^2 + \tfrac{D}{16}h^4 + \mathcal{O}(h^6).$$

Cancel the leading error. Seek $a\,\Phi^{\mathrm{c}}(h/2) + b\,\Phi^{\mathrm{c}}(h)$ with

$$a + b = 1 \quad \text{(consistency)}, \qquad a \cdot \tfrac{1}{4} + b \cdot 1 = 0 \quad \text{(kill the } h^2 \text{ term)} \;\Rightarrow\; a = \tfrac{4}{3},\; b = -\tfrac{1}{3}.$$

This is the standard $p = 2$ Richardson combination

$$\boxed{\Phi^{\mathrm{R}} \;=\; \frac{4\,\Phi^{\mathrm{c}}(h/2) - \Phi^{\mathrm{c}}(h)}{3}.}$$

Appendix — Richardson: error order and use in the hybrid

Residual after cancellation. The h^2 term is gone by construction; the surviving h^4 term is

$$\Phi^{\mathrm{R}} - \Phi = D\big(\tfrac{4}{3}\cdot\tfrac{1}{16} - \tfrac{1}{3}\big)\,h^4 = D\left(\tfrac{1}{12} - \tfrac{4}{12}\right)h^4 = -\tfrac{1}{4}D\,h^4 \;=\; \mathcal{O}(h^4),$$

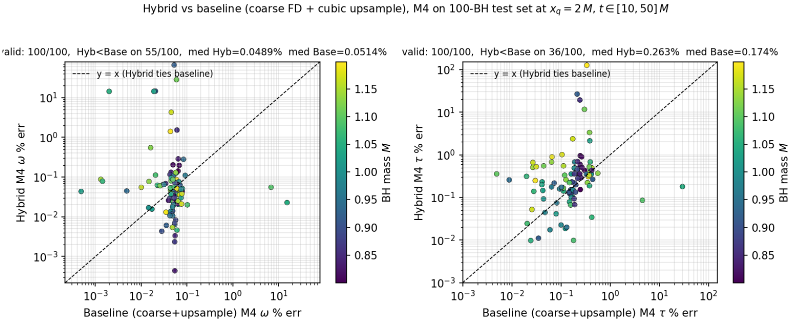
so two *2nd-order* solves combine into a *4th-order* accurate field — without ever computing a finer or fine-FD solve.

Why this is the training target. The network learns $\delta\Phi_{\mathrm{R}} = \Phi^{\mathrm{R}} - \mathcal{U}\Phi^{\mathrm{c}}$, a *label-free* estimate of the coarse-prior discretisation error $\Phi^{\mathrm{f}} - \mathcal{U}\Phi^{\mathrm{c}}$, which it matches to $\mathcal{O}(h^4)$; both vanish as $k \to 1$. The fine-FD field never enters the loss.

Assumptions (and how we honour them).

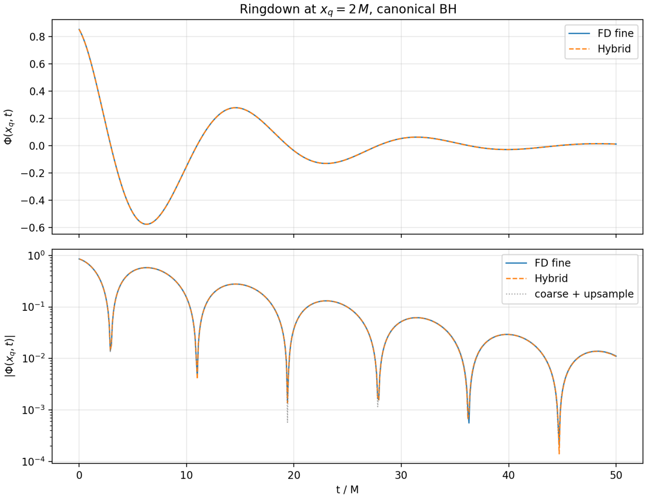
- *Asymptotic regime* — both coarse solves resolve the same smooth solution (verified: Richardson cleans the QNM to $<0.5\%$; an aliased prior could not).
- *Same C* — identical scheme and problem at the two spacings.
- *Upsampling must not contaminate* — the quintic spline $\mathcal{U}$ has $\mathcal{O}(h^6)$ interpolation error, safely below the $\mathcal{O}(h^4)$ signal.

Appendix — hybrid population QNM scatter



Hybrid vs coarse-prior M4 error per black hole ($x_q = 2M$), coloured by mass. The bulk clusters at the joint extractor floor; outliers are ill-conditioned single-window fits that appear on the fine FD reference too.

Appendix — hybrid ringdown at the observer



$\Phi(x_q\!=\!2, t)$: fine FD (blue), hybrid (orange dashed), coarse-up (grey dotted) overlay to $\lesssim 10^{-3}$ across $t \in [0, 50]M$.